

Vivek Mange

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Summary

Robotics Software Engineer with 4+ years of experience designing and deploying autonomous systems across underwater, field, and indoor environments. Specialized in perception, multi-sensor fusion, and real-time robotics systems, with proven success deploying on embedded platforms with measurable improvements in latency, accuracy, and system reliability. Experienced in ROS2-based distributed systems, edge AI deployment, and multi-modal perception for real-world robotic applications.

Professional Experience

University of Maryland, College Park - *Robotics Engineer (Faculty Specialist)* Jan 2025 – Present

- Designed and deployed a real-time perception and decision pipeline integrating classifiers and visual-language models (LLaVA) for autonomous casualty assessment in dynamic environments
- Developed a Kalman filter-based multi-sensor fusion framework integrating RTK-GPS, RGB, thermal, ToF, and radar data, improving localization and target detection accuracy in complex field conditions
- ROS2-based distributed systems with APIs for real-time data reporting in DARPA Triage Challenge
- Developed QT GUI tools for live telemetry, confidence visualization, and debugging workflows
- Collaborated across perception, controls, and systems teams to integrate end-to-end autonomous decision pipelines in real-world deployments

University of Delaware - *Graduate Research Assistant* Jun 2023 – Dec 2024

- Deployed OpenVINS for visual-inertial odometry and implemented multi-sensor fusion for state estimation
- Implemented nonlinear control strategies for stable trajectory tracking in simulation & real-world environments
- Built ROS2-based perception pipeline using YOLOv10 achieving 0.657 mAP@50 at 9.1 Hz on edge devices
- Optimized models for edge deployment on embedded platforms (Jetson), balancing inference speed & accuracy

Swift Robotics, UK - *Robotics Software Engineer* Sep 2022 – Jan 2023

- Developed and optimized LiDAR and depth camera based SLAM pipelines (Cartographer, Hector) and fine-tuned ROS planners (TEB, DWA) for smooth trajectory planning and robust obstacle avoidance
- Deployed indoor Autonomous Mobile Robots (AMRs) with integrated navigation and perception stacks, improving compute efficiency by 30% and reducing system latency by 45%

Drishti Works, India - *Robotics Software Engineer* May 2020 – Aug 2022

- Designed and deployed autonomous mobile robots with end-to-end perception, mapping, and navigation stacks
- Developed real-time diagnostics framework (Sensors, ROS topics, metrics) improving debugging efficiency
- Built Gazebo simulation environments, reducing real-world testing effort and improving deployment by 60%

Core Skills

Robotics: SLAM, Localization, Motion Planning, State Estimation, Git, Linux, Gazebo

Perception: Computer Vision, Multi-Modal Perception, YOLO, OpenCV, PCL

Systems: ROS/ROS2, Distributed Systems, Sensor Fusion, Real-Time Systems, Docker

Programming: Python, C++, PyTorch, MATLAB

Hardware: LiDAR, RGB-D Cameras, IMU, DVL, Jetson, Raspberry Pi, Odroid

Academic Projects

ARM Robot Controller (KUKA LBR iiwa)

- Developed high-precision motion control system achieving 1 mm accuracy using inverse kinematics, Jacobian control, and singularity handling

Autonomous Underwater Vehicle (TMRT)

- Led a team of 30+ engineers to design and deploy an Autonomous Underwater Vehicle (AUV)
- Engineered PID-based control system using IMU and pressure sensors for stable underwater navigation

Publications

- IEEE ICRA: “ODYSSEE – Oyster Detection using Edge-Based Sensor Systems” ([Link](#))
- IRJET: “Autonomous Underwater Vehicle – Electronics and Software Implementation” ([Link](#))

Education

M.S. Robotics, University of Delaware, United States May 2024

B.Tech Electronics & Telecommunications, K.J. Somaiya College of Engineering, India Oct 2020

Honors

- Demonstrated robotics systems to the Prime Minister of India
- TEDx Speaker on Marine Robotics and Autonomous Systems ([Link](#))